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## The fixed- $\sigma$ slices of the residue-dependent solvability obstruction $o_\nu$ : additivity, $B^2$ -killing, and the torsor dichotomy

Corresponds to work-in-progress commit fc3d742 ([github.com/scotmacbeth/work-in-progress](https://github.com/scotmacbeth/work-in-progress)).

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Rick — here are the fixed- $\sigma$  slices you asked for (your email of 2026-09-29, targets (a) and (b)), presented against your two read-points. The underlying object is the obstruction we settled on 09-29; this note slices it by the circle action  $\sigma$  and reports the per-slice tables. Trust grades are stated inline and kept honest: the *slice tables* are **computed** (verification), the *main theorem* they instantiate is **proved**, and the general *torsor* claim remains **computed** (not yet proved in general).

### 1 Setup and what is already proved

Fix a good triplet  $(\nu, \mu, \sigma)$  for a skew-brace extension  $0 \rightarrow D \rightarrow G \rightarrow H \rightarrow 0$  with  $D$  an ideal:  $\nu : (H, \circ) \rightarrow \text{Aut}(D, +)$  a homomorphism (the residue channel),  $\mu : (H, +) \rightarrow \text{Aut}(D, +)$  (trivial here, as  $(G, +)$  is abelian), and  $\sigma : (H, \circ) \rightarrow \text{Aut}(D, +)$  an anti-homomorphism (the circle action). Reading Rathee–Yadav (3.10) as an affine-linear equation in the multiplicative datum for fixed  $\beta$  gives two  $\mathbb{Z}$ -linear operators  $P, Q : C^2 \rightarrow C^3$ ; with  $T(\bar{\tau})(h_1, h_2) = \nu_{h_1 \circ h_2}(\bar{\tau}(h_1, h_2))$  and  $A := P \circ T$  restricted to  $Z_\circ^2 := Z_{\text{Gp}}^2((H, \circ), D; \sigma)$ , the obstruction is

$$o_\nu : H_+^2 \longrightarrow C^3 / A(Z_\circ^2(\sigma)), \quad o_\nu([\beta]) = [Q\beta].$$

**Proved (09-29).** In the trivial-brace-kernel (RY) regime, for general  $H$ :  $o_\nu$  is a well-defined group homomorphism that kills  $B_+^2$ , and

$\text{im } \varphi^+ = \ker o_\nu$

— an additive class  $[\beta]$  extends to a full skew-brace cocycle inducing  $(\nu, \mu, \sigma)$  iff  $o_\nu([\beta]) = 0$ . This is **proved** (structural; the linchpin identity  $P(T d_\circ \theta_1) = Q(d_+ \theta)$  is a formal consequence of  $B^2 \subseteq Z^2$ , exhaustively machine-checked;  $\ker o_\nu = \text{im } \varphi^+$  cross-validated against an independent regular-subgroup enumeration). Full statement and proof: `proofs/2026-09-29-solvability-obstruction-o-nu.tex`, Theorem 1.

**The fixed- $\sigma$  slice framing (your Remark 4).** The target  $A(Z_\circ^2(\sigma))$  depends on  $\sigma$ , while the domain  $H_+^2$  (computed with  $\mu$  trivial) does not. So it is natural to slice: fix  $\sigma$ , vary the admissible residue  $\nu$ , and read  $o_\nu \upharpoonright_\sigma$  as a map  $H_+^2 \rightarrow C^3 / A(Z_\circ^2(\sigma))$ . Your two read-points then become per-slice questions, answered below.

### 2 Read-point (a): each slice is additive and kills $B_+^2$

**Claim (a).** For every fixed  $\sigma$  and admissible  $\nu$ , the slice  $o_\nu \upharpoonright_\sigma$  is an *additive* map  $H_+^2 \rightarrow C^3 / A(Z_\circ^2(\sigma))$  that *kills*  $B_+^2$ ; hence  $\ker o_\nu \upharpoonright_\sigma = \text{im } \varphi^+$  is a based *subgroup* of  $H_+^2$ . This is the slice restriction of Theorem 1 (**proved**); the tables below are the **computed** verification.

I verified, per slice, (i)  $o_\nu(\beta_1 + \beta_2) \equiv o_\nu(\beta_1) + o_\nu(\beta_2) \pmod{A}$  over *all* pairs  $\beta_1, \beta_2 \in Z_+^2$ , and (ii)  $o_\nu(d_+ \theta) \equiv 0$  over *all* additive coboundaries  $d_+ \theta$ . Both pass with no exceptions (`scratch/2026-09-30-o-nu-fix`).

| family                                      | slice                | residue           | additivity | $B^2$ -killing | $ \ker / H_+^2 $ | basepoint |
|---------------------------------------------|----------------------|-------------------|------------|----------------|------------------|-----------|
| $ G =8$<br>$H=\mathbb{Z}/2, D=\mathbb{Z}/4$ | $\sigma = \text{id}$ | $\nu = \text{id}$ | 16/16      | 4/4            | 2/2              | present   |
|                                             | $\sigma = \text{id}$ | $\nu = -1$        | 16/16      | 4/4            | 2/2              | present   |
|                                             | $\sigma = -1$        | $\nu = \text{id}$ | 16/16      | 4/4            | 1/2              | present   |
|                                             | $\sigma = -1$        | $\nu = -1$        | 16/16      | 4/4            | 2/2              | present   |
| $ G =9$<br>$H=\mathbb{Z}/3, D=\mathbb{Z}/3$ | $\sigma = \text{id}$ | $\nu = \text{id}$ | 81/81      | 9/9            | 3/3              | present   |
|                                             | $\sigma = \times 2$  | $\nu = \text{id}$ | 81/81      | 9/9            | 1/3              | present   |

Table 1: Trivial-kernel (RY) slices. Additivity checked over all  $|Z_+^2|^2$  pairs;  $B^2$ -killing over all  $|B_+^2|$  coboundaries. Every slice is a homomorphism killing  $B_+^2$ ; the kernel is a subgroup and always contains the origin (basepoint present). **computed**.

Two remarks tying this to the closed forms. First, the domain size is  $\sigma$ -independent ( $|H_+^2| = 2$  at  $|G| = 8$ ,  $= 3$  at  $|G| = 9$ ), while the kernel *size* moves with the slice: at  $|G| = 8$ ,  $\sigma = -1$ ,  $\nu = \text{id}$  the kernel drops to  $\{[\beta] = 0\}$  (only the split class survives), recovering “ $\nu = \text{id}$  forces  $[\beta] = 0$ ”. Second, on the  $H = \mathbb{Z}/2$  slices the abstract kernel agrees with the closed form

$$o_\nu(b) = 0 \iff (1 + \nu)b \in 2D^\sigma,$$

verified as “MATCHES” on every row — so the  $|G| = 8$  condition  $(1 + \nu)b \in 2D^\sigma$  falls out of the slice exactly as the per- $\sigma$  membership test.

### 3 Read-point (b): the torsor dichotomy

**Claim (b).** A *non-trivial-brace* kernel makes  $o_\nu \upharpoonright_\sigma$  *affine* rather than linear: the realising equation acquires a constant offset  $c$ , so the realisable locus  $\text{im } \varphi^+$  is a *coset* of  $\ker o_\nu$  — a torsor. When that offset is nonzero the coset misses the origin: *no split* ( $[\beta] = 0$ ) *brace exists*, i.e. the torsor has **no basepoint**. This is **computed** on the  $|G| = 16$  witness family and **open** for a general proof.

**The two per- $\sigma$  conditions.** In the trivial-kernel regime (§2) the per- $\sigma$  condition is the *linear/subgroup* membership  $(1 + \nu)b \in 2D^\sigma$ ; the origin always solves it, so the basepoint is always present. In the non-trivial-kernel family  $D = \mathbb{Z}/8$ ,  $\lambda_x^D = \times 5^x$ ,  $L = \{1, 5\} \subsetneq \text{Aut}(D)$ , the residue re-enters through  $\lambda_d$  on the complement and the per- $\sigma$  condition becomes the *affine* three-way lock

$$4b \equiv 2(\nu\sigma - 1) \pmod{8}, \quad \text{offset } c := 2(\nu\sigma - 1) \pmod{8},$$

constant in  $b$ . The realisable  $b$  form the coset  $\{b : 4b = c\}$  of  $\ker(\times 4) = \{0, 2, 4, 6\}$ , and

$$\text{basepoint present} \iff 0 \in \{b : 4b = c\} \iff c = 0 \iff \nu\sigma \equiv 1 \pmod{4}.$$

I verified both the affine lock and this basepoint dichotomy on all 24 genuine order-16 braces with this kernel, sliced by  $\sigma$ .

The dichotomy is crisp: whenever the internal datum  $\nu\sigma$  leaves the socle-like subgroup ( $\nu\sigma \not\equiv 1 \pmod{4}$ ), the offset  $c = 4 \neq 0$  and  $\text{im } \varphi^+$  is a torsor over  $\ker o_\nu$  missing the origin —  $[\beta] = 0$  is *forbidden*, the split additive brace does not exist for that  $(\nu, \sigma)$ . This is the honest replacement of the old “internal-replacement coordinate” story, and the affine offset  $c$  is the intrinsic obstruction to a basepoint.

| slice               | residue          | realisable $b$ | offset $c = 2(\nu\sigma - 1)$ | $\nu\sigma \bmod 4$ | basepoint     |
|---------------------|------------------|----------------|-------------------------------|---------------------|---------------|
| $\sigma = \times 1$ | $\nu = \times 1$ | $\{0\}$        | 0                             | 1                   | present       |
| $\sigma = \times 1$ | $\nu = \times 3$ | $\{1\}$        | 4                             | 3                   | <b>absent</b> |
| $\sigma = \times 1$ | $\nu = \times 5$ | $\{0\}$        | 0                             | 1                   | present       |
| $\sigma = \times 1$ | $\nu = \times 7$ | $\{1\}$        | 4                             | 3                   | <b>absent</b> |
| $\sigma = \times 3$ | $\nu = \times 3$ | $\{0\}$        | 0                             | 1                   | present       |
| $\sigma = \times 3$ | $\nu = \times 7$ | $\{0\}$        | 0                             | 1                   | present       |
| $\sigma = \times 5$ | $\nu = \times 1$ | $\{0\}$        | 0                             | 1                   | present       |
| $\sigma = \times 5$ | $\nu = \times 3$ | $\{1\}$        | 4                             | 3                   | <b>absent</b> |
| $\sigma = \times 5$ | $\nu = \times 5$ | $\{0\}$        | 0                             | 1                   | present       |
| $\sigma = \times 5$ | $\nu = \times 7$ | $\{1\}$        | 4                             | 3                   | <b>absent</b> |
| $\sigma = \times 7$ | $\nu = \times 3$ | $\{0\}$        | 0                             | 1                   | present       |
| $\sigma = \times 7$ | $\nu = \times 7$ | $\{0\}$        | 0                             | 1                   | present       |

Table 2: Non-trivial-kernel ( $|G| = 16$ ,  $D = \mathbb{Z}/8$ ,  $L = \{1, 5\}$ ) slices. In each slice the affine lock  $4b \equiv c$  holds on every realising brace, and the basepoint is present iff  $c = 0$  iff  $\nu\sigma \equiv 1 \pmod{4}$ . The rows with  $\nu\sigma \equiv 3$  ( $c = 4$ ) realise *only*  $[\beta] \neq 0$ : a torsor with no basepoint. **computed**.

**Grade, honestly.** The affine/torsor structure and the basepoint dichotomy are **computed** on this family and match every genuine brace; a general- $H$ , general-kernel proof that  $\text{im } \varphi^+$  is an  $o_\nu$ -affine torsor (with  $c$  identified intrinsically as the class of the  $\lambda_d$ -complement cochain) is the natural next target and is **open**. Nothing in the slice computation contradicts the 09-29 proof file — the linear-slice tables of §2 and the affine-slice table of §3 reproduce its  $|G| = 8$  and  $|G| = 16$  conditions exactly.

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Verification script: `scratch/2026-09-30-o-nu-fixed-sigma-slices.py` (reuses the four 09-29 scripts). All slices: additivity and  $B^2$ -killing pass with no exceptions; the  $|G| = 16$  affine lock and basepoint dichotomy pass on all 24 genuine braces. Happy to co-develop the general torsor proof if you want to take the offset-as-a-class route.